



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO ANNAI FUEL SERVICE

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

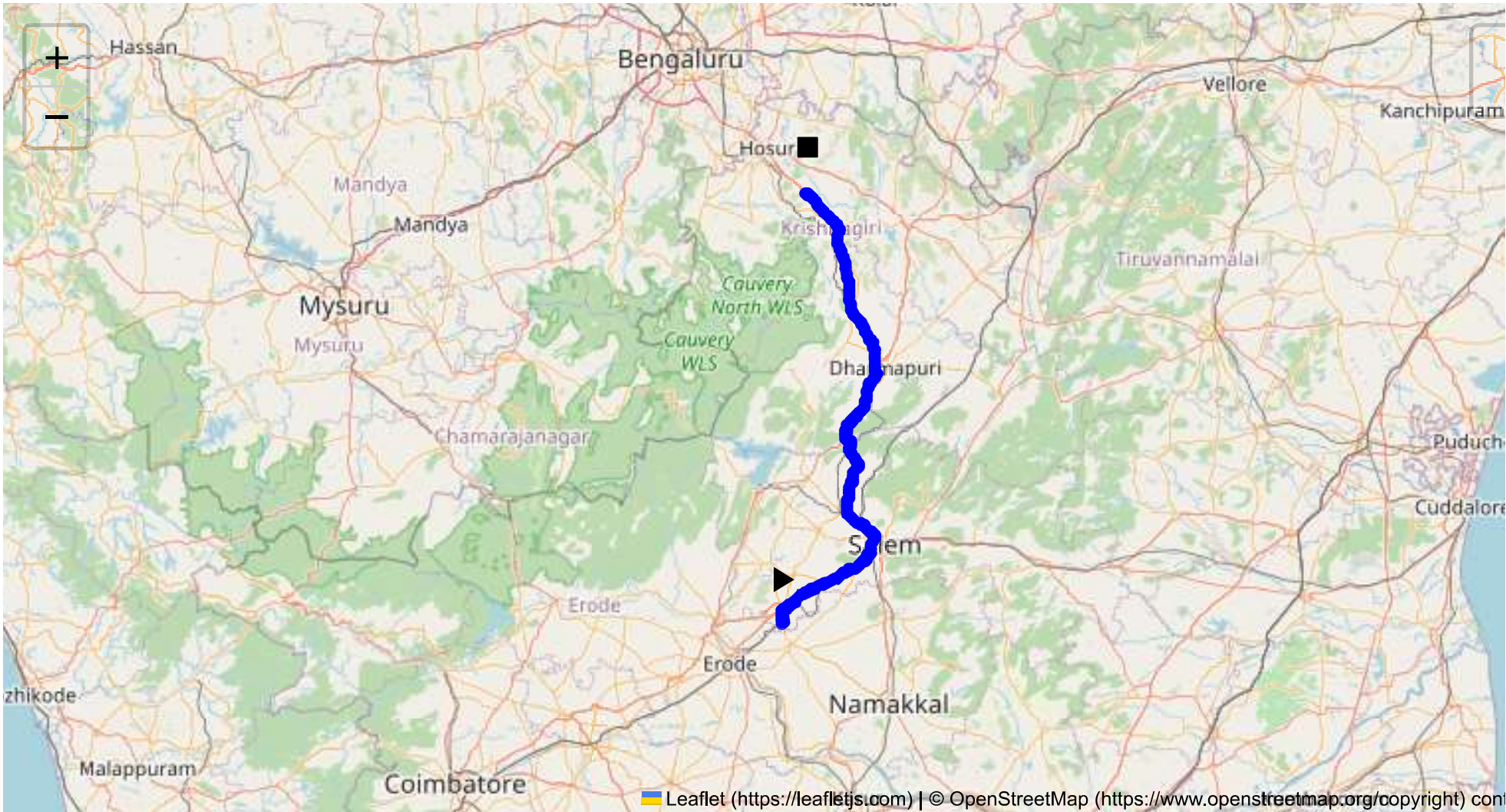
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 165.52 km
Estimated Duration: 3.1 hours
Adjusted Duration (Heavy Vehicle): 3.9 hours
Start: (11.4381, 77.8734)
End: (12.611051, 77.935165)



Welcome to the Journey Risk Management Study

Route Overview

The route from CVQF+23W, Sangagiri, Tamil Nadu 637301, India to JW6P+84Q, Uddanapalli, Tamil Nadu 635119, India covers approximately 165.52 kilometers. This route likely follows national highways and state roads commonly used for both local and regional transportation needs. The journey typically takes about 3.10 hours for heavy vehicles carrying hazardous materials.

Weather Conditions

- **Typical Weather:** The region experiences a tropical climate with hot summers (March to June), monsoons (June to September), and mild winters (November to February).
- **Weather Hazards:** Monsoon season may bring heavy rains, leading to waterlogging and potential flooding in low-lying areas. Drivers should be cautious of slippery roads and reduced visibility during heavy rainfall.

Traffic Patterns

- **Peak Hours:** Urban areas near the start and end points may experience congestion during typical commuting hours (8:00 AM - 10:00 AM and 5:00 PM - 8:00 PM).
- **Congestion-Prone Areas:** Expect traffic slowdowns near larger towns and cities, including Erode and Krishnagiri, especially at junctions with busy local roads.

Road Quality and Infrastructure

- **Road Quality:** Main highways are generally well-maintained, but secondary roads may have potholes, especially after the rainy season. Some rural roads might be narrower and require careful navigation.
- **Infrastructure:** Adequate infrastructure exists with petrol stations and rest stops along the highway. However, services may become sparse in more rural segments.

Alternative Routes for Emergencies

- Consider NH44 as an alternative for portions of the journey to bypass potentially hazardous or congested areas. It's important to have updated maps for real-time detours in case of unexpected closures or traffic.

Local Regulations for Hazardous Material Transport

- **Regulations:** Heavy vehicles must adhere to local and national transport regulations, including obtaining necessary permits for hazardous materials. Restrictions on driving through densely populated areas during peak hours might apply.
- **Requirements:** Drivers should be familiar with documentation requirements and carry material safety data sheets (MSDS) for all hazardous materials.

Historical Incidents

- **Past Incidents:** Highways in Tamil Nadu have seen accidents involving heavy vehicles, often due to speeding, driver fatigue, or poor road conditions. It is crucial to maintain safe speeds and vigilance at all times.

Environmental Considerations

- **Sensitive Areas:** Sections of the route may pass near agricultural lands and water bodies, requiring extra care to avoid contamination risks from hazardous materials.
- **Wildlife Crossings:** Be alert for animal crossings in rural and forested areas, particularly in the evening and night.

Communication Coverage

- **Dead Zones:** Expect communication challenges in remote stretches between major towns; ensure up-to-date maps and alternative communication means are available.
- **Network Providers:** Major providers generally offer good coverage, but verify with local operators for specific blind spots.

Emergency Response Times

- **Urban Areas:** Faster response times (under 30 minutes) near major towns and highways.
- **Rural Areas:** Response may be delayed (up to an hour or more) in less accessible areas; having a plan for emergency measures is essential.

Overall Summary of Risk Assessment

The route from Sangagiri to Uddanapalli involves both urban and rural stretches requiring heightened awareness of weather conditions, road quality, and traffic patterns. Ensure compliance with all transportation regulations for hazardous materials and prepare for emergencies with backup plans and alternative routes. Finally, maintain proactive communication and risk mitigation strategies to ensure a safe and efficient journey.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	High	11.44900, 77.87406	15 KM/Hr
3	Turn	High	11.45028, 77.87202	15 KM/Hr
4	Turn	Medium	11.46400, 77.87323	30 KM/Hr
5	Turn	High	11.47403, 77.86841	15 KM/Hr
6	Turn	High	11.49149, 77.89601	15 KM/Hr
7	Turn	High	11.49224, 77.89601	15 KM/Hr
8	Turn	Medium	11.53512, 77.95717	30 KM/Hr
9	Turn	Medium	11.53606, 77.95869	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
10	Turn	Medium	11.62403, 78.10893	30 KM/Hr
11	Turn	High	11.64183, 78.11810	15 KM/Hr
12	Turn	Medium	11.64338, 78.11998	30 KM/Hr
13	Turn	Medium	11.73968, 78.05098	30 KM/Hr
14	Turn	Medium	11.86790, 78.08347	30 KM/Hr
15	Turn	Medium	11.91357, 78.05868	30 KM/Hr
16	Turn	Medium	11.92556, 78.06499	30 KM/Hr
17	Turn	Medium	11.93694, 78.05425	30 KM/Hr
18	Turn	Medium	11.95531, 78.04698	30 KM/Hr
19	Turn	High	11.95803, 78.04964	15 KM/Hr
20	Turn	Medium	11.96068, 78.04960	30 KM/Hr
21	Turn	High	11.96370, 78.05142	15 KM/Hr
22	Turn	High	11.96542, 78.04938	15 KM/Hr
23	Turn	High	11.96650, 78.04927	15 KM/Hr
24	Turn	Medium	12.12036, 78.13284	30 KM/Hr
25	Turn	Medium	12.50466, 78.03128	30 KM/Hr
26	Blind Spot	Blind Spot	12.51373, 78.03208	10 KM/Hr
27	Turn	Medium	12.51382, 78.02925	30 KM/Hr
28	Turn	Medium	12.51628, 78.02574	30 KM/Hr
29	Turn	Medium	12.52905, 78.02044	30 KM/Hr
30	Turn	Medium	12.61032, 77.93611	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
1	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
2	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
3	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
4	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
5	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
6	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
7	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
9	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
10	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
11	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
12	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
13	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
15	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
20	hospital	Bharathi Hospital, Salem	11.6492647, 78.1204131	30 km/h	Medium
21	hospital	Sisu Hospital Inida Ltd	11.6505276, 78.1205003	30 km/h	Medium
22	hospital	Aishwarya Hospital	11.6716291, 78.1267807	30 km/h	Medium
24	hospital	C.M. Hospital	11.6747345, 78.1255034	30 km/h	Medium
25	pharmacy	Healthy Pharmacy	11.6762714, 78.127744	30 km/h	Medium
26	pharmacy	Thulasi Pharmacy	11.6761991, 78.1277377	30 km/h	Medium
27	hospital	Sri Gokulam Speciality Hospital	11.6788724, 78.1274258	30 km/h	Medium
28	hospital	Global Medical Center	11.6867049, 78.1186697	30 km/h	Medium
29	hospital	Global Medical Centre Jagirammalayam	11.6894409, 78.1108847	30 km/h	Medium
30	hospital	Manipal Hospital	11.703072, 78.1026612	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
34	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
35	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
37	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
39	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
40	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
41	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
14	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
16	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
17	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
18	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
19	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
23	college	Sona College	11.6768, 78.1257052	30 km/h	Medium
31	college	Govt Engineering College	11.7128537, 78.0877084	30 km/h	Medium
32	university	Periyar University	11.7186758, 78.0771089	30 km/h	Medium
33	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
36	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium